

AQI TOWER

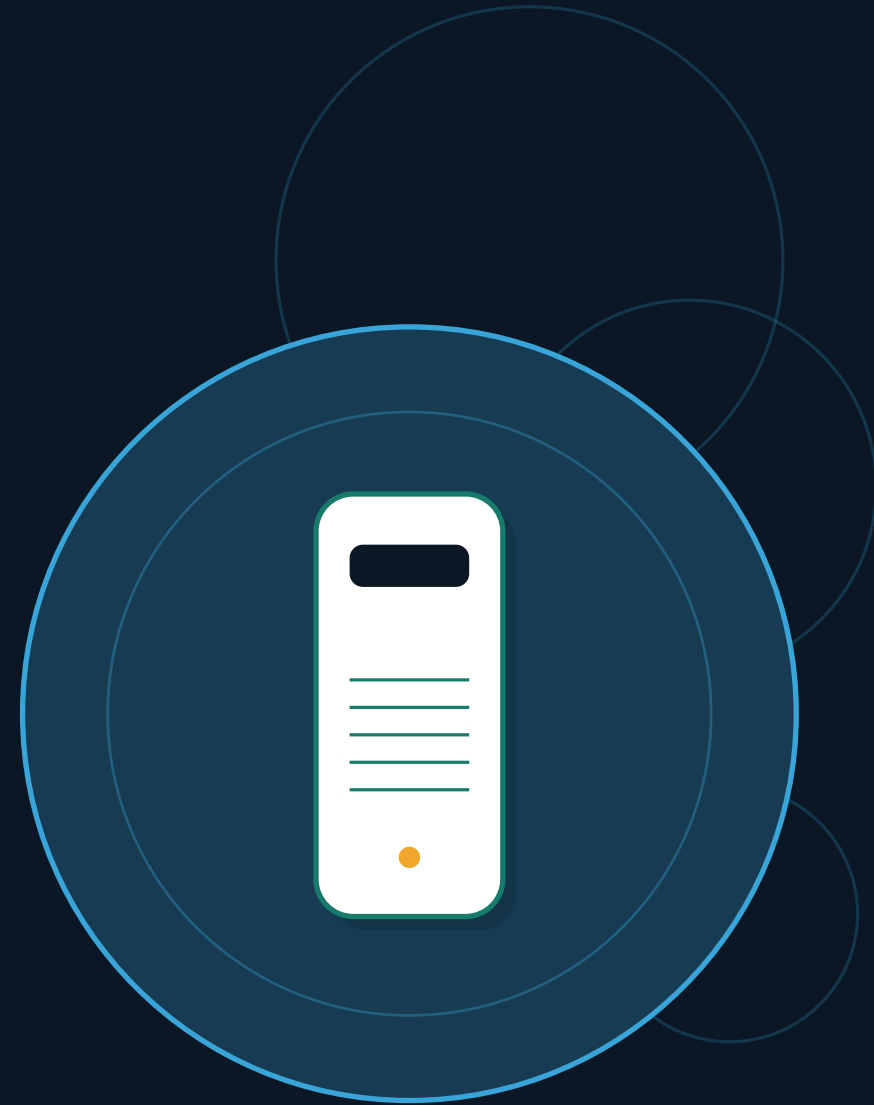
One tower. One clean-air bubble.
Later, a connected chain of bubbles.

DIGITAL PROTOTYPE

EVIDENCE FIRST

CURRENT STAGE

Digital design and simulation completed so far. Physical prototype not built yet.



PROTOTYPE COVERAGE IDEA - NOT YET MEASURED

Executive Summary

A one-page overview for people seeing the AQI Tower for the first time.



Vision for 2026

- Move from a strong digital design toward expert review and buying test parts.
- Begin controlled physical testing before making performance claims.
- Prepare the path from one tested unit to connected clean-air bubbles.



Agenda-driven meetings held

Not available in the technical project records. Add the verified number from the team meeting register.

PROGRESS

Work completed

- Built an adjustable 3D computer model.
- Completed 10 airflow simulation cases.
- Used real fan data and studied an H13 HEPA filter.
- Found and corrected an air shortcut around the filter.
- Studied particle paths, special carbon material and formaldehyde.

DIRECTION

Key decisions taken

- Use the Improved Airflow Shape found by computer comparison.
- Use a dry washable pre-filter instead of water spray.
- Study formaldehyde using a special carbon material.
- Make no removal or coverage claim without real testing.

Unresolved challenges

- No physical prototype
- Bubble size is unknown
- Official name not selected
- Safe laboratory access

Support required

- Engineering review
- Materials and instruments
- Testing budget and lab

Upcoming month

- Freeze test specifications
- Contact suppliers and experts
- Prepare safe component tests

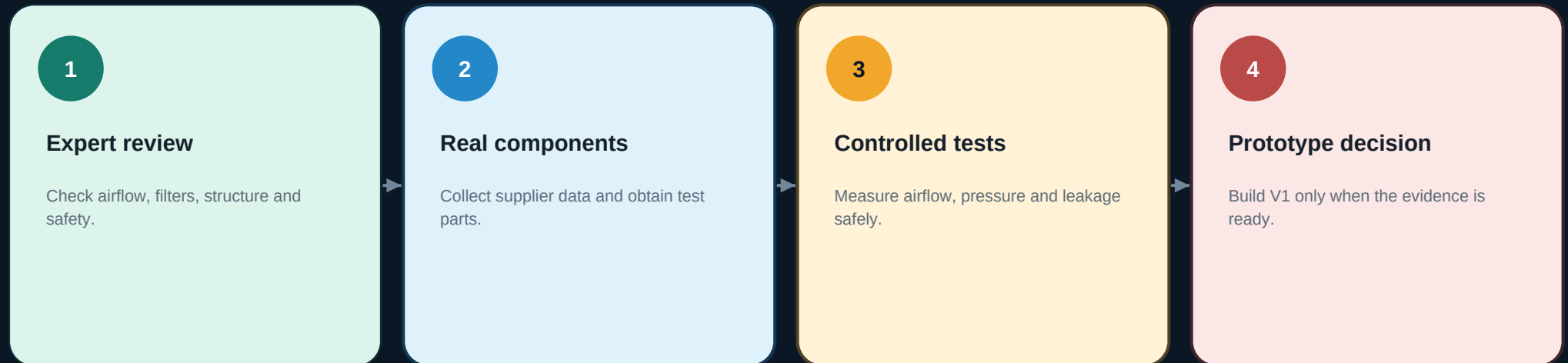
Vision for 2026

Move carefully from digital engineering toward the first controlled physical tests.

THE 2026 GOAL

Turn a strong computer-tested idea into a safe test-ready prototype plan.

The goal is progress toward expert review, parts, component tests and a Prototype V1 decision - not a promise of a finished outdoor product.



Long-term vision

Prove one local clean-air bubble first. Only then study connected towers and a larger network.

One Tower = One Bubble

The bubble is a simple way to show the local area that one unit is intended to support.

PROTOTYPE NOW

Start small and learn



ONE SMALL LOCAL COVERAGE ZONE

TEST AND IMPROVE



FUTURE VISION

Build a chain of bubbles



CONNECTED UNITS CAN SUPPORT A LARGER AREA

Bubble means planned coverage

It is not a sealed dome.

Exact size is still unknown

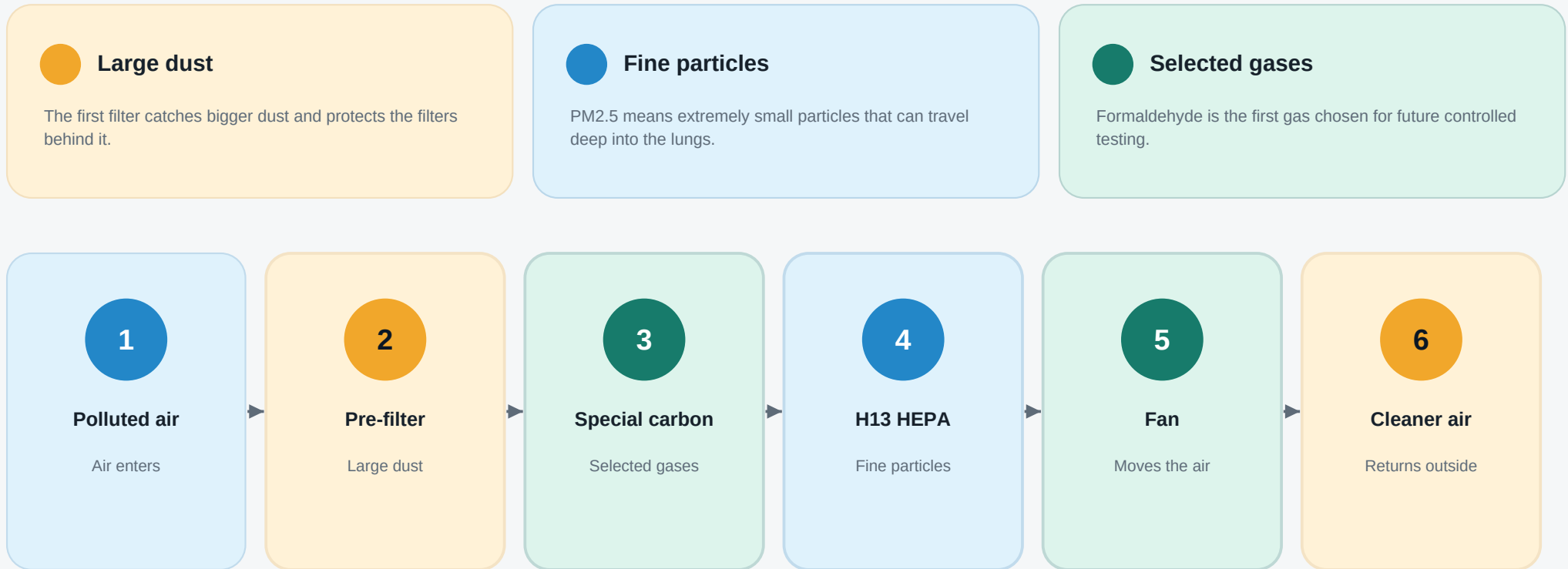
Outdoor tests must measure it.

Growth comes after proof

First validate one small prototype.

The Problem and the Simple Air Path

The project studies dust, fine particles and selected harmful gases.



HOW PROOF WILL WORK

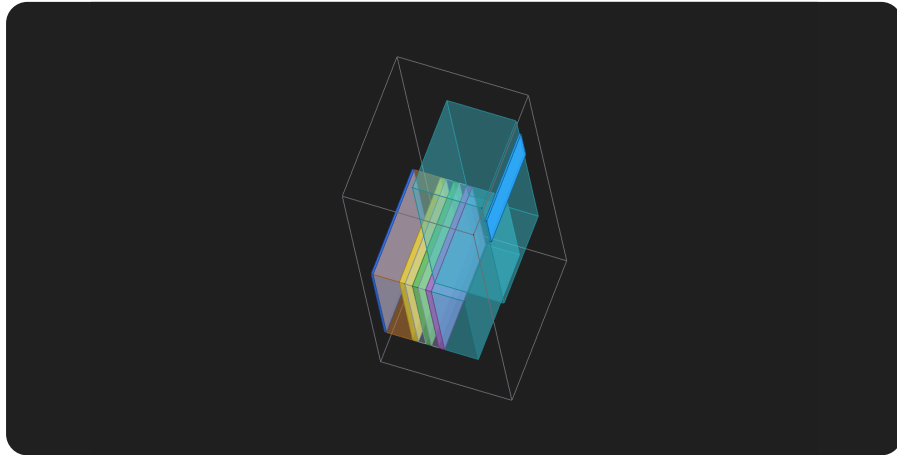
Measure the incoming and outgoing air - never guess.

The indoor air path was developed first. Outdoor bubble performance is the next big validation problem.

Design update: Water spray was studied and rejected. A dry washable pre-filter is recommended. Solar assistance remains a later idea.

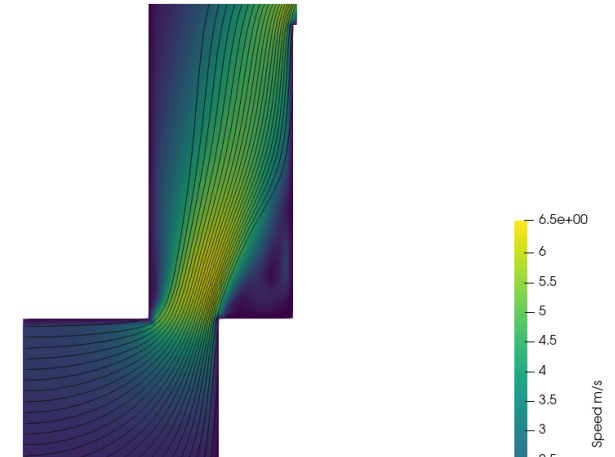
What We Have Done So Far

The team built a digital engineering foundation before spending money on a physical tower.



DIGITAL MODEL

We built an adjustable computer model of the tower so the shape can be changed without starting again.



AIRFLOW SIMULATION

We completed 10 airflow cases and used real manufacturer fan data instead of a made-up fan value.

COMPLETED PROGRESS



Plan

Purpose and requirements



Model

First CAD concept



Simulate

Airflow and fan



Improve

Better path and sealed filter



Study

Particles and gas material



Decide

Dry pre-filter, no water

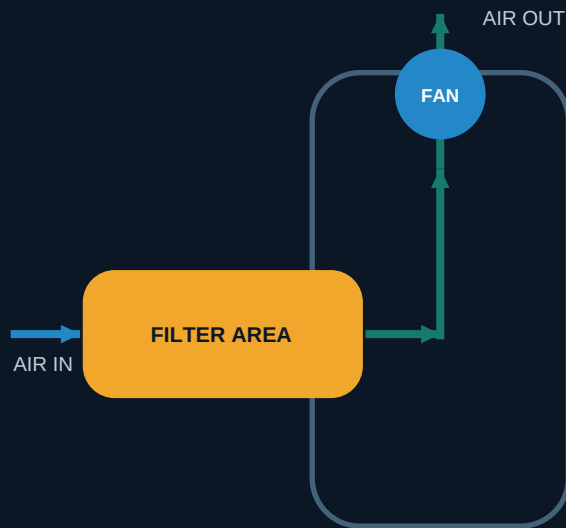
The Improved Airflow Shape

The engineering files call it GEO_C. In this report, we use the simple name instead.

WHAT IT IS

A smoother path inside the tower

Earlier shapes made some air turn backwards or circle inside. This version gives the air a cleaner turn toward the top fan.

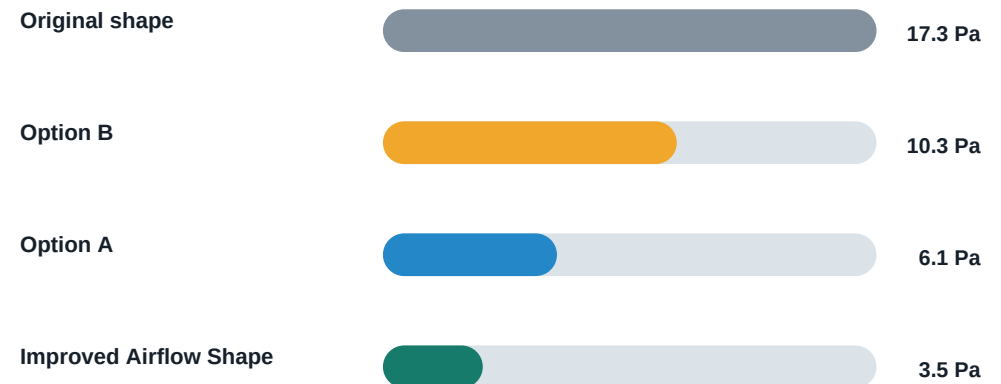


It is the best current computer-tested internal shape. It is not the final manufactured tower.

WHY IT WAS SELECTED

Less resistance for the fan

Pressure loss means how hard the fan must work to push air through the empty tower shape.



About 5x lower pressure loss

Computer simulation only. The shape has not been built or physically tested.

Plain-language rule: Say 'Improved Airflow Shape'. Mention GEO_C only when someone needs to find the engineering file.

What the Computer Studies Found

These numbers guide the prototype. None is a physical product result.

1,496 m3/h

Empty-tower airflow

Predicted airflow before the main filter resistance is added.

1,332 m3/h

Airflow with H13

Predicted airflow after adding the sealed HEPA filter model.

111 Pa

Estimated HEPA airflow resistance

Computer simulation estimate of how much the fine-particle filter slows the air.

45% to ~0%

Bypass corrected

A digital sealing plate removed the modelled shortcut around the filter.

PARTICLE PATH STUDY

About 62% reached the filter plane

The computer released 224 particles in each size group and followed their paths. Reaching the filter does not mean the particle was captured or removed.

ARRIVAL IS NOT REMOVAL

Real filter capture must be measured.

WHAT THE H13 LABEL MEANS

The selected filter candidate is manufacturer-rated H13. That rating belongs to the filter under standard test conditions - it does not prove the complete AQI Tower removes 99.95% of outdoor pollution.

What We Have Decided

These choices guide the next tests. They are not promises about a finished product.

1 Simulation before fabrication

Learn from the computer model before spending heavily on hardware.

2 Use real fan data

The project uses a six-point manufacturer fan curve, not a fixed guess.

3 Use the improved shape

It gave the lowest simulated empty-tower pressure loss of the tested options.

4 Dry filters, not water spray

Use a washable pre-filter and H13 HEPA candidate. The water stage was rejected.

5 One gas target first

Formaldehyde is the first gas to study with activated-carbon candidates.

6 Measured data before gas CFD

Do not invent carbon performance. Test pressure loss and adsorption first.

Material note

Activated carbon is documented in the current project. Biochar has not yet been verified or selected, so the two names should not be used as if they mean the same thing.

Questions We Still Need to Answer

These are the main gaps between a strong digital study and a proven physical product.

1 Physical proof

No tower has been built. Real airflow, noise, power and leakage are unknown.

2 Outdoor bubble

The effect of wind, tower spacing and the real clean-air area is unknown.

3 Filter performance

Real PM2.5 removal and full-system filter sealing must be measured.

4 Gas treatment

Formaldehyde removal, carbon lifetime and humidity effects are not measured.

5 Name and product design

The official project/device name is not selected. The final hardware design is also not frozen.

6 Safety and approval

Electrical safety, laboratory approval and Indian certification need expert review.

Why this matters

Answering these questions protects the team from making claims too early and shows exactly where expert help is needed.

Who Can Help Move the Project Forward

The next stage needs validation support before large-scale manufacturing support.



Simple 4-Week Plan

This is a proposed plan based on the current stage, not a fixed promise.

WEEK 1

Freeze specifications

- Choose the official project/device name.
- Shortlist the dry G4/M5 pre-filter.
- Request real pressure data.
- Send carbon supplier enquiries.

WEEK 2

Professional review

- Review CAD and airflow work.
- Check fan, filters and sealing.
- Clarify the first outdoor use case.

WEEK 3

Prepare component tests

- Finalize quotations and parts.
- Prepare the carbon pressure rig.
- Write the safety test procedure.

WEEK 4

Prototype readiness

- Freeze only the minimum test design.
- Prepare airflow, leakage and power checks.
- Decide if Prototype V1 is ready.



Safety gate

Do not start formaldehyde gas testing until a qualified laboratory, safe containment, reference measurement and formal approval are ready.

What Is Not Proven Yet

The AQI Tower is a digital engineering project, not yet a validated air purifier.

CURRENT LIMITATIONS

- Physical prototypes built: 0
- Real airflow and pressure: not measured
- PM2.5 and formaldehyde removal: not measured
- Clean Air Delivery Rate: not established
- Outdoor bubble size: unknown
- Multi-tower field test: not done

CLAIMS WE MUST NOT MAKE YET

No invented radius.

No invented coverage area.

No invented removal percentage.

These values can be stated only after controlled physical and outdoor tests.

THE RESPONSIBLE PATH FORWARD



BUILD ONE - MEASURE HONESTLY - IMPROVE - THEN SCALE